

AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Nitrogen dioxide (no2)
Observed / expected baseline	17.9 / 2.08494 ppb
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-16 02:00 UTC
Location	29.6957, -95.4992
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 617	57.73 to 100; mean 89.408	94.24 at 2026-06-16 01:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 617	22.7778 to 33.3333; mean 26.6362	26 at 2026-06-16 01:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 617	0 to 360; mean 119.514	100 at 2026-06-16 01:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 617	0 to 12.3467; mean 2.7198	3.08666 at 2026-06-16 01:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	10 / 120	5850.83 to 5926.85; mean 5886.95	5885.85 at 2026-06-16 00:00Z; 2 h before event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	10 / 120	32.0322 to 1802.53; mean 343.786	104.919 at 2026-06-16 00:00Z; 2 h before event
noaa_gfs	Precipitable water (precipitable_water)	mm	10 / 120	50.8211 to 69.5435; mean 60.0142	60.7863 at 2026-06-16 00:00Z; 2 h before event
noaa_gfs	Surface pressure (surface_pressure)	hPa	10 / 120	999.209 to 1013.9; mean 1008.08	1008.11 at 2026-06-16 00:00Z; 2 h before event
noaa_gfs	850-hPa temperature (t_850)	degC	10 / 120	16.785 to 19.498; mean 18.2799	18.9384 at 2026-06-16 00:00Z; 2 h before event
noaa_gfs	10-m eastward wind (u_10m)	m/s	10 / 120	-3.72304 to 1.69078; mean -0.8794	-0.419447 at 2026-06-16 00:00Z; 2 h before event
noaa_gfs	10-m northward wind (v_10m)	m/s	10 / 120	-1.42555 to 5.19652; mean 1.8155	0.124454 at 2026-06-16 00:00Z; 2 h before event
openaq	Ozone (ozone)	ppm	17 / 1109	-0.001 to 0.032; mean 0.0136	0.009 at 2026-06-16 02:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 212	5 to 110; mean 24.4906	27 at 2026-06-16 02:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	90 / 3143	-2.6 to 52; mean 6.7883	9.1 at 2026-06-16 02:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 355	0 to 100; mean 92.062	100 at 2026-06-16 02:01Z; 2 min after event
openweather	Relative humidity (humidity)	percent	5 / 355	68 to 100; mean 90.7493	93 at 2026-06-16 02:01Z; 2 min after event
openweather	Precipitation rate (precipitation)	mm/h	5 / 355	0 to 24.39; mean 1.2864	0 at 2026-06-16 02:01Z; 2 min after event
openweather	Surface pressure (pressure)	hPa	5 / 355	1005 to 1016; mean 1011.16	1012 at 2026-06-16 02:01Z; 2 min after event
openweather	Air temperature (temperature)	degC	5 / 355	23.79 to 33.3; mean 26.7107	25.55 at 2026-06-16 02:01Z; 2 min after event
openweather	Wind direction (wind_direction)	degree	5 / 355	0 to 358; mean 158.062	173 at 2026-06-16 02:01Z; 2 min after event
openweather	Wind speed (wind_speed)	m/s	5 / 355	0.07 to 5.67; mean 2.0495	1.11 at 2026-06-16 02:01Z; 2 min after event
purpleair	PM2.5 (pm25)	ug/m3	67 / 4829	0 to 46.2; mean 8.0664	15.2 at 2026-06-16 02:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 6 h 50 min before event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	6 / 6	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 6 h 50 min before event
sentinel5p	CO column (s5p_co_column)	mol/m^2	1 / 1	0.0259566 to 0.0259566; mean 0.026	0.0259566 at 2026-06-14 19:24Z; 30 h 35 min before event
sentinel5p	CO granules available (s5p_co_granule_available)	count	7 / 7	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 6 h 50 min before event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	1 / 1	0.000111617 to 0.000111617; mean 0.0001	0.000111617 at 2026-06-14 19:24Z; 30 h 35 min before event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 6 h 50 min before event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	1 / 1	2.03572e-05 to 2.03572e-05; mean 0	2.03572e-05 at 2026-06-14 19:24Z; 30 h 35 min before event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 6 h 50 min before event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 6 h 50 min before event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	1 / 1	-6.10873e-06 to -6.10873e-06; mean -0	-6.10873e-06 at 2026-06-14 19:24Z; 30 h 35 min before event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-06-15 19:09Z; 6 h 50 min before event
tceq	Carbon monoxide (co)	ppm	3 / 213	0.1 to 1.1; mean 0.2225	0.5 at 2026-06-16 02:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	13 / 858	-2.1 to 27; mean 5.5844	17.9 at 2026-06-16 02:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 219	-0.6 to 0.6; mean -0.1137	0.4 at 2026-06-16 02:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-14 12Z 78.62, 18Z 69.98, 06-15 00Z 88.16, 06Z 92.9, 12Z 93.04, 18Z 90.65, 06-16 00Z 93.26, 06Z 97.13, 12Z 93.15, 18Z 86.17, 06-17 00Z 89.65, 06Z 95.39, 12Z 94.45
asos	Air temperature (temperature)	06-14 12Z 29.11, 18Z 30.74, 06-15 00Z 27.65, 06Z 26.9, 12Z 25.98, 18Z 26.07, 06-16 00Z 25.96, 06Z 25.46, 12Z 25.49, 18Z 26.98, 06-17 00Z 26.19, 06Z 24.8, 12Z 24.22
asos	Wind direction (wind_direction)	06-14 12Z 153.1, 18Z 176.3, 06-15 00Z 115, 06Z 124, 12Z 173.5, 18Z 106.5, 06-16 00Z 68.46, 06Z 67.17, 12Z 155.8, 18Z 106.6, 06-17 00Z 106, 06Z 101.3, 12Z 109.4
asos	Wind speed (wind_speed)	06-14 12Z 4.001, 18Z 4.066, 06-15 00Z 2.229, 06Z 2.379, 12Z 2.744, 18Z 1.271, 06-16 00Z 1.533, 06Z 1.126, 12Z 3.255, 18Z 2.048, 06-17 00Z 2.815, 06Z 3.921, 12Z 7.259
noaa_gfs	500-hPa geopotential height (gh_500)	06-14 18Z 5926, 06-15 00Z 5921, 06Z 5922, 12Z 5902, 18Z 5893, 06-16 00Z 5886, 06Z 5890, 12Z 5872, 18Z 5865, 06-17 00Z 5855, 06Z 5859, 12Z 5852
noaa_gfs	Planetary boundary layer height (pbl_height)	06-14 18Z 1472, 06-15 00Z 458.9, 06Z 224.7, 12Z 239.3, 18Z 270.1, 06-16 00Z 155.5, 06Z 66.58, 12Z 143.6, 18Z 425.1, 06-17 00Z 296.2, 06Z 123.2, 12Z 249.9
noaa_gfs	Precipitable water (precipitable_water)	06-14 18Z 58.86, 06-15 00Z 60.51, 06Z 60.03, 12Z 58.78, 18Z 65.4, 06-16 00Z 61.28, 06Z 62.55, 12Z 64.15, 18Z 63.08, 06-17 00Z 55.38, 06Z 55.85, 12Z 54.29
noaa_gfs	Surface pressure (surface_pressure)	06-14 18Z 1011, 06-15 00Z 1010, 06Z 1011, 12Z 1011, 18Z 1011, 06-16 00Z 1008, 06Z 1009, 12Z 1008, 18Z 1008, 06-17 00Z 1005, 06Z 1005, 12Z 1002
noaa_gfs	850-hPa temperature (t_850)	06-14 18Z 17.9, 06-15 00Z 18.87, 06Z 19.26, 12Z 18.51, 18Z 17.22, 06-16 00Z 18.82, 06Z 18.41, 12Z 17.78, 18Z 17.54, 06-17 00Z 18.81, 06Z 17.98, 12Z 18.27
noaa_gfs	10-m eastward wind (u_10m)	06-14 18Z -0.8708, 06-15 00Z -0.9255, 06Z -0.5358, 12Z -0.4235, 18Z -0.9461, 06-16 00Z -0.4284, 06Z -0.6977, 12Z 0.5038, 18Z 0.3331, 06-17 00Z -1.673, 06Z -2.254, 12Z -2.634
noaa_gfs	10-m northward wind (v_10m)	06-14 18Z 4.358, 06-15 00Z 2.858, 06Z 2.473, 12Z 1.891, 18Z 2.389, 06-16 00Z -0.2305, 06Z 1.215, 12Z 1.724, 18Z 4.105, 06-17 00Z 0.5973, 06Z 0.5837, 12Z -0.1755
openaq	Ozone (ozone)	06-14 12Z 0.01829, 18Z 0.02207, 06-15 00Z 0.0138, 06Z 0.0103, 12Z 0.00878, 18Z 0.01526, 06-16 00Z 0.007596, 06Z 0.002412, 12Z 0.009808, 18Z 0.01683, 06-17 00Z 0.01458, 06Z 0.0189, 12Z 0.02033
openaq	PM10 (pm10)	06-14 12Z 32.44, 18Z 22.17, 06-15 00Z 33.06, 06Z 49.44, 12Z 41.22, 18Z 12.22, 06-16 00Z 20.89, 06Z 23.69, 12Z 18.33, 18Z 15, 06-17 00Z 18.89, 06Z 14.44, 12Z 14.22
openaq	PM2.5 (pm25)	06-14 12Z 6.551, 18Z 8.402, 06-15 00Z 9.636, 06Z 9.642, 12Z 7.469, 18Z 4.56, 06-16 00Z 7.97, 06Z 7.125, 12Z 4.252, 18Z 2.61, 06-17 00Z 2.806, 06Z 4.772, 12Z 5.027
openweather	Cloud cover (cloud_cover)	06-14 12Z 62.67, 18Z 81.5, 06-15 00Z 95.83, 06Z 99.86, 12Z 99.13, 18Z 99.93, 06-16 00Z 99.57, 06Z 99.33, 12Z 99.9, 18Z 100, 06-17 00Z 96.81, 06Z 75.69, 12Z 35.91
openweather	Relative humidity (humidity)	06-14 12Z 81.6, 18Z 73.03, 06-15 00Z 89, 06Z 92.72, 12Z 93.48, 18Z 94.13, 06-16 00Z 92.93, 06Z 95.23, 12Z 94.17, 18Z 91.79, 06-17 00Z 91.1, 06Z 93.86, 12Z 94.73
openweather	Precipitation rate (precipitation)	06-14 12Z 0.2573, 18Z 0.1917, 06-15 00Z 0.203, 06Z 0.19, 12Z 3.307, 18Z 4.336, 06-16 00Z 0.7973, 06Z 1.585, 12Z 1.259, 18Z 0.7517, 06-17 00Z 0.4219, 06Z 0.7976, 12Z 3.236
openweather	Surface pressure (pressure)	06-14 12Z 1015, 18Z 1013, 06-15 00Z 1013, 06Z 1013, 12Z 1014, 18Z 1012, 06-16 00Z 1012, 06Z 1011, 12Z 1011, 18Z 1009, 06-17 00Z 1008, 06Z 1006, 12Z 1006
openweather	Air temperature (temperature)	06-14 12Z 29.77, 18Z 31.85, 06-15 00Z 27.99, 06Z 27.09, 12Z 26.29, 18Z 25.43, 06-16 00Z 25.95, 06Z 25.36, 12Z 25.52, 18Z 26.41, 06-17 00Z 26.26, 06Z 25.04, 12Z 24.17
openweather	Wind direction (wind_direction)	06-14 12Z 182.3, 18Z 182, 06-15 00Z 151.6, 06Z 164.7, 12Z 187.4, 18Z 163.7, 06-16 00Z 125.5, 06Z 198.5, 12Z 184.4, 18Z 165, 06-17 00Z 114.1, 06Z 111.2, 12Z 97.73
openweather	Wind speed (wind_speed)	06-14 12Z 2.644, 18Z 3.247, 06-15 00Z 2.106, 06Z 2.097, 12Z 1.986, 18Z 1.815, 06-16 00Z 1.037, 06Z 1.571, 12Z 1.688, 18Z 2.229, 06-17 00Z 2.293, 06Z 2.014, 12Z 2.502

Source	Metric	Values
purpleair	PM2.5 (pm25)	06-14 12Z 7.781, 18Z 10.93, 06-15 00Z 11.81, 06Z 10.79, 12Z 7.792, 18Z 6.415, 06-16 00Z 9.367, 06Z 9.342, 12Z 4.713, 18Z 5.178, 06-17 00Z 4.927, 06Z 7.278, 12Z 8.673
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	CO granules available (s5p_co_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	NO2 granules available (s5p_no2_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
sentinel5p	SO2 granules available (s5p_so2_granule_available)	06-14 18Z 1, 06-15 18Z 1, 06-16 18Z 1
tceq	Carbon monoxide (co)	06-14 12Z 0.2167, 18Z 0.1933, 06-15 00Z 0.2167, 06Z 0.1778, 12Z 0.2333, 18Z 0.2667, 06-16 00Z 0.2778, 06Z 0.2278, 12Z 0.2556, 18Z 0.2444, 06-17 00Z 0.2167, 06Z 0.1556, 12Z 0.1889
tceq	Nitrogen dioxide (no2)	06-14 12Z 2.03, 18Z 1.408, 06-15 00Z 2.617, 06Z 2.791, 12Z 5.438, 18Z 7.302, 06-16 00Z 10.67, 06Z 8.838, 12Z 5.771, 18Z 6.854, 06-17 00Z 6.909, 06Z 4.976, 12Z 4.862
tceq	Sulfur dioxide (so2)	06-14 12Z -0.2417, 18Z -0.1278, 06-15 00Z -0.1833, 06Z -0.2389, 12Z -0.2111, 18Z -0.07222, 06-16 00Z 0.01667, 06Z -0.09444, 12Z -0.08333, 18Z -0.07222, 06-17 00Z 0.005556, 06Z -0.1, 12Z -0.1222

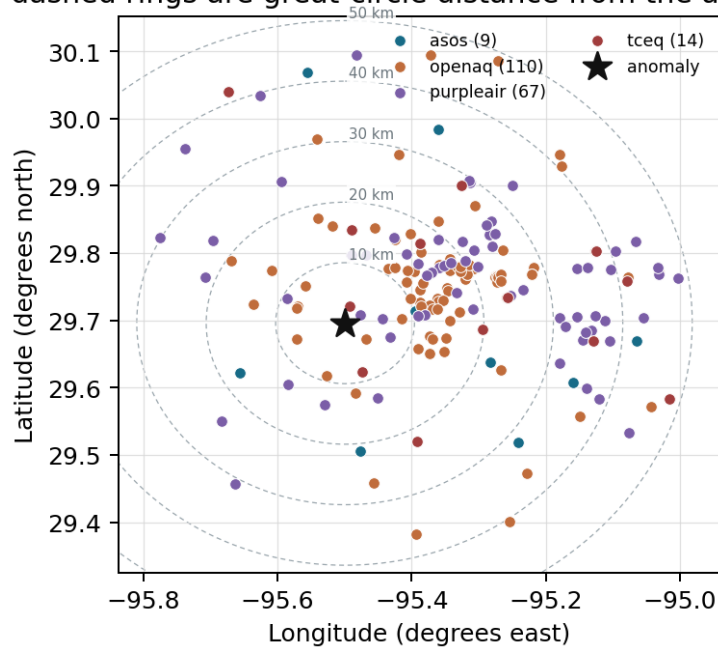
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

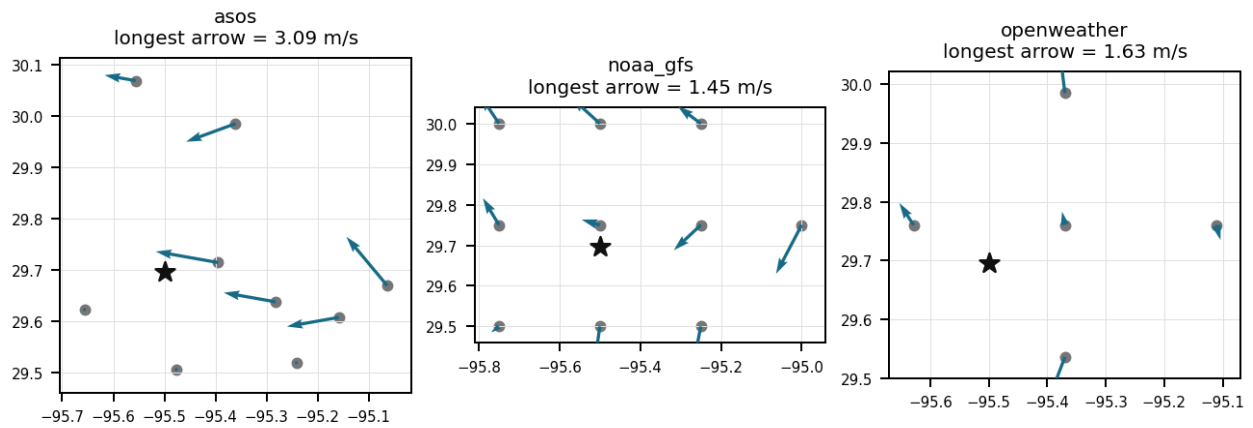
Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (10.269 km) 92.98, SGR (17.253 km) 92.37, AXH (21.198 km) 86.27, HOU (21.928 km) 87.87, LVJ (31.726 km) 87.65, EFD (34.344 km) 87.83, IAH (34.767 km) 91.68, DWH (41.756 km) 89.48, +1 more
asos	Air temperature (temperature)	MCJ (10.269 km) 26.19, SGR (17.253 km) 26.46, AXH (21.198 km) 26.56, HOU (21.928 km) 26.99, LVJ (31.726 km) 26.86, EFD (34.344 km) 27.1, IAH (34.767 km) 26.35, DWH (41.756 km) 26.33, +1 more
asos	Wind direction (wind_direction)	MCJ (10.269 km) 124.6, SGR (17.253 km) 125.8, AXH (21.198 km) 64.03, HOU (21.928 km) 135.7, LVJ (31.726 km) 121.4, EFD (34.344 km) 104.9, IAH (34.767 km) 134.4, DWH (41.756 km) 111.5, +1 more
asos	Wind speed (wind_speed)	MCJ (10.269 km) 3.915, SGR (17.253 km) 2.779, AXH (21.198 km) 1.186, HOU (21.928 km) 3.101, LVJ (31.726 km) 2.408, EFD (34.344 km) 3.551, IAH (34.767 km) 2.508, DWH (41.756 km) 1.822, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.50 (6.033 km) 5887, gfs:29.50,-95.50 (21.766 km) 5888, gfs:29.75,-95.25 (24.811 km) 5887, gfs:29.75,-95.75 (24.957 km) 5887, gfs:29.50,-95.25 (32.471 km) 5888, gfs:29.50,-95.75 (32.583 km) 5888, gfs:30.00,-95.50 (33.831 km) 5887, gfs:30.00,-95.25 (41.501 km) 5887, +2 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.50 (6.033 km) 380.9, gfs:29.50,-95.50 (21.766 km) 305.9, gfs:29.75,-95.25 (24.811 km) 359.3, gfs:29.75,-95.75 (24.957 km) 302.2, gfs:29.50,-95.25 (32.471 km) 339.8, gfs:29.50,-95.75 (32.583 km) 263.5, gfs:30.00,-95.50 (33.831 km) 413.8, gfs:30.00,-95.25 (41.501 km) 369.1, +2 more
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.50 (6.033 km) 59.84, gfs:29.50,-95.50 (21.766 km) 60.48, gfs:29.75,-95.25 (24.811 km) 60.6, gfs:29.75,-95.75 (24.957 km) 59.75, gfs:29.50,-95.25 (32.471 km) 60.64, gfs:29.50,-95.75 (32.583 km) 60.39, gfs:30.00,-95.50 (33.831 km) 59.2, gfs:30.00,-95.25 (41.501 km) 59.87, +2 more

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.50 (6.033 km) 1008, gfs:29.50,-95.50 (21.766 km) 1009, gfs:29.75,-95.25 (24.811 km) 1009, gfs:29.75,-95.75 (24.957 km) 1007, gfs:29.50,-95.25 (32.471 km) 1010, gfs:29.50,-95.75 (32.583 km) 1008, gfs:30.00,-95.50 (33.831 km) 1007, gfs:30.00,-95.25 (41.501 km) 1008, +2 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.50 (6.033 km) 18.25, gfs:29.50,-95.50 (21.766 km) 18.26, gfs:29.75,-95.25 (24.811 km) 18.28, gfs:29.75,-95.75 (24.957 km) 18.3, gfs:29.50,-95.25 (32.471 km) 18.36, gfs:29.50,-95.75 (32.583 km) 18.3, gfs:30.00,-95.50 (33.831 km) 18.25, gfs:30.00,-95.25 (41.501 km) 18.27, +2 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.50 (6.033 km) -1.006, gfs:29.50,-95.50 (21.766 km) -1.27, gfs:29.75,-95.25 (24.811 km) -0.7402, gfs:29.75,-95.75 (24.957 km) -0.8594, gfs:29.50,-95.25 (32.471 km) -1.209, gfs:29.50,-95.75 (32.583 km) -1.177, gfs:30.00,-95.50 (33.831 km) -0.7202, gfs:30.00,-95.25 (41.501 km) -0.6894, +2 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.50 (6.033 km) 1.62, gfs:29.50,-95.50 (21.766 km) 1.832, gfs:29.75,-95.25 (24.811 km) 1.617, gfs:29.75,-95.75 (24.957 km) 1.718, gfs:29.50,-95.25 (32.471 km) 2.181, gfs:29.50,-95.75 (32.583 km) 1.647, gfs:30.00,-95.50 (33.831 km) 1.817, gfs:30.00,-95.25 (41.501 km) 1.857, +2 more
openaq	Ozone (ozone)	311 (0.006 km) 0.01486, 312 (8.347 km) 0.01459, 1319333 (13.167 km) 0.01546, 287 (13.542 km) 0.01453, 319 (15.426 km) 0.009773, 325 (19.781 km) 0.01348, 3055 (22.069 km) 0.01419, 3378 (23.726 km) 0.009877, +9 more
openaq	PM10 (pm10)	15852419 (19.314 km) 26.71, 13380082 (23.726 km) 26.8, 271 (35.926 km) 19.9
openaq	PM2.5 (pm25)	2071339 (0.006 km) 9.561, 8967119 (3.902 km) 9.081, 15588192 (7.397 km) 6.965, 13955816 (7.399 km) 7.525, 15588193 (7.44 km) 7.411, 15588190 (7.49 km) 7.181, 8967001 (8.223 km) 6.714, 14140748 (8.44 km) 6.437, +82 more
openweather	Cloud cover (cloud_cover)	grid:west (14.405 km) 93.55, grid:center (14.417 km) 92.24, grid:south (21.745 km) 93.18, grid:north (34.501 km) 88.51, grid:east (38.16 km) 92.83
openweather	Relative humidity (humidity)	grid:west (14.405 km) 91.2, grid:center (14.417 km) 91.94, grid:south (21.745 km) 88.37, grid:north (34.501 km) 91.44, grid:east (38.16 km) 90.8
openweather	Precipitation rate (precipitation)	grid:west (14.405 km) 1.222, grid:center (14.417 km) 1.167, grid:south (21.745 km) 1.494, grid:north (34.501 km) 1.123, grid:east (38.16 km) 1.425
openweather	Surface pressure (pressure)	grid:west (14.405 km) 1011, grid:center (14.417 km) 1011, grid:south (21.745 km) 1011, grid:north (34.501 km) 1011, grid:east (38.16 km) 1011
openweather	Air temperature (temperature)	grid:west (14.405 km) 26.49, grid:center (14.417 km) 26.73, grid:south (21.745 km) 26.82, grid:north (34.501 km) 26.56, grid:east (38.16 km) 26.95
openweather	Wind direction (wind_direction)	grid:west (14.405 km) 157.6, grid:center (14.417 km) 154.2, grid:south (21.745 km) 154.2, grid:north (34.501 km) 159.3, grid:east (38.16 km) 165
openweather	Wind speed (wind_speed)	grid:west (14.405 km) 1.456, grid:center (14.417 km) 1.585, grid:south (21.745 km) 2.531, grid:north (34.501 km) 2.042, grid:east (38.16 km) 2.634
purpleair	PM2.5 (pm25)	186315 (2.576 km) 10.2, 46237 (5.405 km) 0.2603, 166451 (6.9 km) 10, 28851 (9.308 km) 10.34, 293735 (10.612 km) 10.42, 51959 (11.334 km) 8.873, 293725 (11.536 km) 10.27, 301969 (11.717 km) 11.71, +59 more
tceq	Carbon monoxide (co)	482011052 (17.021 km) 0.04043, 482011035 (23.713 km) 0.1352, 482011039 (35.927 km) 0.1319
tceq	Nitrogen dioxide (no2)	482010055 (0.0 km) 5.358, 482011066 (2.946 km) 7.533, 482010047 (15.427 km) 8.601, 482011052 (17.021 km) 15.3, 482010416 (19.781 km) 5.671, 480391004 (22.053 km) 2.27, 482011035 (23.713 km) 8.792, 482010024 (28.285 km) 1.842, +5 more
tceq	Sulfur dioxide (so2)	482010051 (8.334 km) 0.04932, 482011035 (23.713 km) -0.4164, 482011039 (35.927 km) 0.02603

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 35

The PM2.5 levels in the affected area exceeded the normal range, indicating a significant increase in particulate matter in the air.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 2 of 35

Regional TCEQ monitors recorded elevated network-wide 6-hour mean NO2 levels peaking at 10.67 ppb leading into the 00:00 UTC period on June 16, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 3 of 35

TCEQ surface observations and GFS atmospheric model data indicate that a drop in planetary boundary layer height to below 160 meters near 02:00 UTC restricted vertical mixing, causing localized concentration of surface NO2 emissions.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 4 of 35

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 5 of 35

The tceq NO2 monitor at 29.696, -95.499 recorded 17.9 ppb at 2026-06-16T02:00:00+00:00, compared with an expected value of 2.084939759036145 ppb, corresponding to a z-score of 8.059392672370368 and severe anomaly classification.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 6 of 35

The PM2.5 levels are elevated at 15.2 ug/m3, which is higher than the mean of 8.0664 ug/m3.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 7 of 35

Ozone titration by freshly emitted NO is a likely physical cause of the observed chemical signature because ozone near the anomaly dropped to 0.009 ppm and the broader 6-hour ozone mean declined further after 00 UTC while the TCEQ network 6-hour NO2 mean rose to 10.67 ppb, indicating nighttime conversion within a fresh urban plume rather than regionally mixed daytime production.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 8 of 35

The TCEQ Houston monitor recorded a severe NO2 spike to 17.9 ppb at 2026-06-16T02:00:00+00:00 while the 6-hour network mean had already risen to 10.67 ppb for 06-16 00Z, indicating elevated local nighttime NOx conditions around the anomaly.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 9 of 35

A shallow nocturnal planetary boundary layer and light surface winds suppressed atmospheric mixing, trapping surface nitrogen dioxide concentrations near the ground.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 10 of 35**

Air-chemistry context favored a fresh local combustion plume because ozone at the anomaly time was only 0.009 ppm while PM_{2.5} remained modest near 9.1 to 15.2 ug/m³ and SO₂ was only 0.4 ppb, which is inconsistent with a broad regional photochemical or smoke episode.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 11 of 35**

A severe nocturnal nitrogen dioxide concentration spike of 17.9 ppb was detected at 02:00 UTC on June 16, 2026, at the TCEQ monitoring station in Houston.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 12 of 35

Concurrently elevated PM2.5 concentrations observed across PurpleAir and OpenAQ sensor networks support local primary combustion emissions, such as vehicular traffic or nearby industrial activity, as the main driver of the event.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 13 of 35

Accumulation of evening vehicular traffic emissions along nearby urban transport corridors contributed to elevated localized nitrogen dioxide levels under low-wind conditions.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 14 of 35

Meteorological conditions near the anomaly featured weak winds near 1.1 to 3.1 m/s, very high humidity near 93 to 94 percent, complete cloud cover, and a shallow planetary boundary layer near 105 m, which would strongly suppress dispersion and favor NO2 accumulation.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 15 of 35

Low ambient ozone levels recorded near the anomaly location indicate nocturnal titration of ozone by local nitric oxide emissions into nitrogen dioxide.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 16 of 35

Low nighttime ozone concentrations recorded across OpenAQ and TCEQ monitoring networks at 02:00 UTC confirm suppressed photolysis and rapid titration of ozone by fresh nitric oxide emissions under dark conditions.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 17 of 35

Weak dispersion and nighttime trapping are a likely physical cause of the NO₂ enhancement because surface winds near the event were light at about 1.1 to 3.1 m/s, humidity was very high near 93 to 94 percent, and modeled planetary boundary layer height was very low near 105 m around 00 UTC and fell lower afterward, favoring accumulation of locally emitted NO_x.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 18 of 35

The increase in PM2.5 levels is due to an industrial accident at a nearby facility.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 19 of 35

At 2026-06-16T02:00:00+00:00, ozone at the nearest openaq sensor was 0.009 ppm, and the 6-hour mean of tceq NO2 for 2026-06-16 00Z was 10.67 ppb across the local monitor set.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 20 of 35

A combination of stagnant weather patterns and increased emissions from vehicles and industrial activities has contributed to the anomaly.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 21 of 35

The air quality anomaly in Houston, Texas on June 16th was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 22 of 35

A severe nitrogen dioxide (NO₂) anomaly was recorded by TCEQ at coordinates (29.696, -95.499) on June 16, 2026, at 02:00 UTC, reaching a concentration of 17.9 ppb compared to an expected value of 2.08 ppb.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 23 of 35**

A broad regional combustion or secondary aerosol episode is not strongly supported because PM_{2.5} was only modestly elevated near the site, PM₁₀ was not extreme, SO₂ remained near background, and the available satellite context does not show corroborating elevated trace-gas columns close in time to the event.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 24 of 35**

Planetary boundary layer height dropped below 150 meters alongside low surface wind speeds under 1.6 m/s, suppressing atmospheric dispersion in Houston.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 25 of 35

The anomaly occurred at 2026-06-16T02:00:00+00:00 in Houston, Texas at the tceq NO2 monitor 482010055 located at 29.696, -95.499.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 26 of 35**

A nearby fresh combustion NOx plume is a likely physical cause of the 17.9 ppb NO2 spike because the anomalous monitor reading occurred with concurrent low ozone of 0.009 ppm and moderately elevated CO of 0.5 ppm, which is a pattern consistent with recent local traffic, industrial, marine, rail, or other combustion emissions affecting the site.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 27 of 35**

The anomaly occurred on June 16th at approximately 02:00:00+00:00, with the nearest sensor located 2.576 km away.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 28 of 35

Nighttime accumulation under weak dispersion is supported because the anomaly occurred with very high humidity near 94 percent, low local wind speeds near 1.1 to 3.1 m/s, nearly complete cloud cover, and a modeled planetary boundary layer height near 105 m around 00 UTC that fell further to very low 6-hour mean values by 06 UTC, all of which favor trapping of primary pollutants near the surface.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 29 of 35**

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 30 of 35**

The air quality anomaly in Houston, Texas is caused by high levels of NO2 concentrations.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 31 of 35

The CO levels were also elevated, suggesting a potential contribution from nearby industrial activities or vehicle emissions.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 32 of 35**

The air quality anomaly in Houston, Texas is caused by high levels of CO concentrations.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 33 of 35**

A local fresh NO_x emission plume is supported because the anomalous NO₂ concentration of 17.9 ppb at 02:00 UTC coincided with suppressed ozone of 0.009 ppm at the same location and time, which is a classic near-source titration signature of recent NO emissions rather than aged regional pollution.

Mark exactly one:☐ V☐ I☐ U**Note (optional):****Claim 34 of 35**

The anomaly is caused by a high-pressure system that has been lingering over the region.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 35 of 35

Industrial combustion emissions from regional point sources accumulated locally due to weak boundary layer ventilation and stagnant winds.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.